

DIRECTION CHANGES AND BACKLASH IN FDM 3D PRINTED CYCLOIDAL DRIVES

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ABSTRACT

In order to investigate the reliability of cycloidal drives made via readily-accessible FDM 3D printing polymers, five cycloidal drives were FDM 3D Printed from polylactic acid (PLA) and their backlash was recorded over the course of 5 hours of continuous use with a varying number of elapsed direction changes. The backlash in a cycloidal drive is a measure of its positional precision, and it was measured via the phase shift between the input and output shafts when given an oscillating input at a fixed magnitude angular velocity. At all measured points in time over the 5 hour wear-in period, the backlash in every drive was 0.000 ± 0.002 rad. This result indicates that there was no measurable loss in the positional precision in the cycloidal drives over 5 hours of use.

Keywords: Cycloidal Drive, Backlash, PLA Wear, Cycloidal Drive Backlash, FDM Additive Manufacturing

INTRODUCTION

Robotic systems that require high precision and accuracy generally cannot rely on their motors directly for positional control, either because the motors cannot guarantee angular position to the necessary precision or because motors that can produce such precision cannot generate enough torque to be useful on their own [1]. This lack of power or precision is an issue most commonly addressed by using gearboxes to increase motor torque and decrease motor speed. However, most gearboxes on the market are by their nature imprecise, and have output shafts that have a range of free motion even when the input shaft is held stationary [2], [3]. This range of motion is commonly referred to as backlash. To avoid gearbox backlash causing positional imprecision, less conventional options must be considered.

Cycloidal drives are one such option. These drives have low backlash because they do not rely on gears as do other gearbox designs [4]. Cycloidal drives are, therefore, prevalent in robotic systems, and are particularly useful in

hobbyist robotics, where many components are 3D printed via Fused Deposition Modeling (FDM) machines [4], [5]. When compared to competing high-precision gear reduction designs, cycloidal drives have no atypical material requirements, such as flexibility, which allows for more freedom in material selection [6]. This freedom means that custom cycloidal drive components can be straightforward to 3D print and provide good initial performance regardless of the chosen material. The option to manufacture drives in this way, however, raises the question of whether drives made of common FDM 3D printing materials can maintain positional precision over timescales relevant for hobbyist use.

It is, therefore, worth investigating how much influence wear and usage patterns have on this precision, and thus whether the ‘lifespan’ of 3D printed gearboxes is of concern. In this case, usage patterns refer to the frequency of direction changes. These direction changes are notable because the process of rotating through the backlash period of a cycloidal drive may accelerate wear due to impact when the end of the backlash period is hit. In particular, the wear on parts made of polylactic acid (PLA) is worthy of investigation, because PLA is a common 3D printing material for its ease of use in FDM machines and its biodegradability [7], [8].

This wear is investigated by examining five identical 3D printed, PLA cycloidal drives. One control drive is worn in via continuous rotation in one direction over a 5 hour wear-in period, and the other four drives have direction changes happen at varying frequencies over the wear-in period. By comparing the backlash in the drives with direction changes to the control drive before, during, and after the wear-in period, the effect of prolonged use and direction changes on the backlash in 3D printed, PLA cycloidal drives is determined.

CYCLOIDAL DRIVE DESIGN

Cycloidal drives function by using a combination of an eccentric cam shaft and an epitrochoidal disk to generate a gear reduction ratio. An example of an epitrochoid is shown in Figure 1. The input to the drive is connected to the cam shaft, and the cam shaft is connected to the epitrochoidal disk via a bearing. As the eccentric cam rotates, it forces the disk into a ring of rollers. Because of the epitrochoidal shape of the disk, the eccentric rotation of the cam, and the bearing isolating the disk rotation from that of the cam, the disk slowly ‘rolls’ along the rollers in the opposite direction of the input shaft. The output shaft is then connected to the disk via several through-holes. The rollers need not be bearings or free to rotate at all: the rollers can instead be cylindrical pins fused to the outer housing of the drive. This is the case for the drive design used in this study, shown in Figure 2. Commonly, a cycloidal drive has two or more disks oriented out of phase to cancel out any vibrations caused by rotation of the eccentric components.

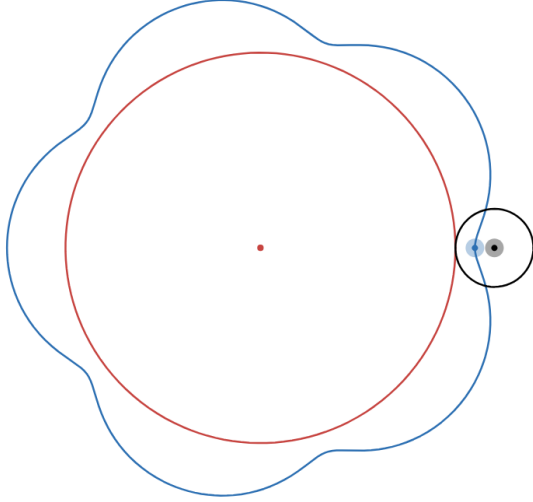


Figure 1: Generation of an epitrochoid shape with five lobes by tracing a point on a circle as it is ‘rolled’ around a larger circle.

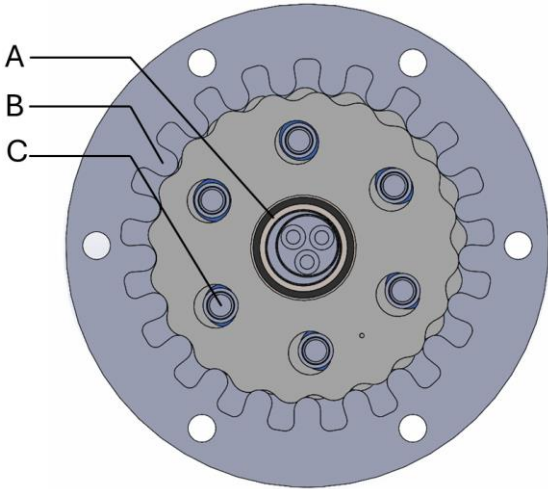


Figure 2: Internal mechanism for a cycloidal drive. The input shaft in the center has two eccentric cams

(with one visible in the figure), with each press fit into a bearing (labelled A). Each bearing is itself press fit into an epitrochoidal disk, shown in tan. There is one more roller pin than there are lobes on each disk. Because of this difference, as the input shaft rotates, the cams force the disks to roll along the ring of roller pins fused with the main body of the drive (labelled B). The output shaft rotates with the disks via six pins (labelled C). This design includes two disks out of phase with each other to cancel out vibrations.

Cycloidal drives have well-defined parameters that are necessary to achieve expected operation. The lobed disks that generate the gear reduction of the cycloidal drive have an exterior contour defined by an epitrochoid, and in most computer aided design programs a convenient way to create an epitrochoidal shape is via parametric equations. The parametric equations in t that define the contour of the cycloidal disks in this study are shown in Equations 1 and 2 [9].

$$x(t) = R \cos(t) - R_r \cos\left(t - \frac{\sin((1-N)t)}{\frac{R}{\varepsilon N} - \cos((1-N)t)}\right) - \varepsilon \cos(Nt) \quad (1)$$

$$y(t) = -R \sin(t) + R_r \sin\left(t - \frac{\sin((1-N)t)}{\frac{R}{\varepsilon N} - \cos((1-N)t)}\right) + \varepsilon \sin(Nt) \quad (2)$$

N is the number of rollers in the design, R_r is the radius of the rollers in millimeters, ε is the eccentricity of the cam/input shaft in millimeters, and R is the radius of the disk in millimeters. The design used in this study has values $N = 21$, $R = 37.4$ mm, $R_r = 3$ mm, and $\varepsilon = 1$ mm. The computer aided design program used for this design is SOLIDWORKS. To create the contour of the disks, t was swept from 0 to $2\pi - 0.001$ and completed with a spline. This is done because SOLIDWORKS does not allow an equation-driven curve to start and end on the same point [9]. The gear reduction ratio of the gearbox is determined by the number of lobes on the disk, $N - 1$. For this cycloidal drive design, this ratio is 20:1. To achieve static balance, two disks are included in the design, 180° out of phase of each other. The output shaft is connected to the cycloidal disks via through holes in the disks, with diameters equal to the diameters of the output shaft pins plus the eccentricity value of the cam shaft [9]. An overview of the full design is in Figure 3.

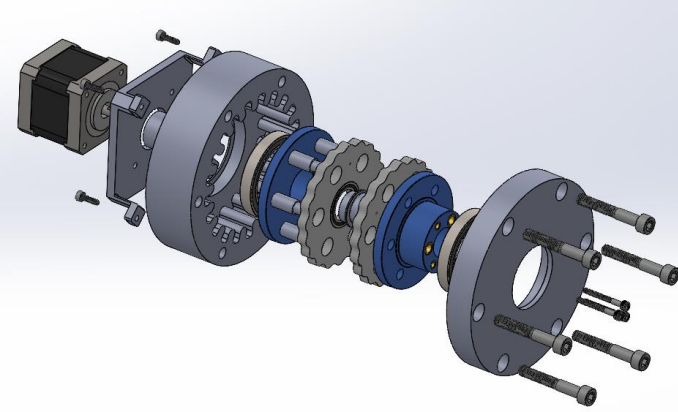


Figure 3: Exploded view of full cycloidal drive design, showing NEMA-17 input stepper motor, cycloidal disks (tan), and output shaft (blue).

CYCLOIDAL DRIVE BACKLASH

Backlash is defined as the angular range in a speed reduction system wherein a gearbox component, commonly an output shaft or internal gear, can rotate freely without full contact with other components in the gearbox. Without full contact, the output shaft is not sufficiently constrained and therefore does not rotate with the input shaft. It can thus be rotated freely by external influence, which introduces imprecision. In other gear reduction methods, the use of gears with involute teeth inherently causes backlash [2], [3]. A benefit of cycloidal drives is that they eschew involute gear tooth shapes, and thus avoid this particular source of backlash. The components of cycloidal drives, however, must be assembled in order to function, and no manufacturing process can produce parts with perfect dimensional accuracy. As such, tolerancing or imprecision in the dimensions of cycloidal drive components re-introduces backlash. Importantly, because tolerances are necessary in manufacturing to account for imprecision, there is no such thing as a cycloidal drive entirely devoid of backlash. The backlash can be low enough to be difficult to measure, but it is always present.

WEAR-IN AND BACKLASH MEASUREMENT OF SAMPLE DRIVES

In order to determine the relationship between cycloidal drive backlash and wear, five sample drives were manufactured via FDM 3D Printing and were worn in over the course of five hours, each with a different number of total direction changes over the wear-in period. To better emulate real-world applications, the drives were lubricated with WD-40 Specialist Dry Lube and the cycloidal disks had the seams caused by the FDM 3D printing process sanded flush to the contour of the disks. The backlash of

each drive was measured before any wear, and was then measured after every elapsed hour of the wear-in period. There were, therefore, two experimental setups used in this experiment: one to wear in the drives, and one to measure the backlash in the drives.

BACKLASH MEASUREMENT

After every hour of wear-in, backlash measurements were taken. The backlash can be measured by recording the time delay between the input and output shafts of the cycloidal drive when under a triangle wave input motion, i.e. constant velocity rotation for a quarter rotation of the output shaft, then a quarter rotation in the other direction at the same magnitude velocity, and so on. Because the input shaft has to rotate through the backlash range of the output shaft, there is a time delay between the input changing directions and the output following suit. If the input oscillates as per a triangle wave, this time delay between the input and output would become a phase shift between the input and output. From this phase shift, and the fact that the input shaft always rotates with a constant magnitude angular velocity, the backlash of the cycloidal drive can be calculated as shown in Equation 3.

$$\theta = \frac{\omega t}{R} \quad (3)$$

θ is the backlash range in the output shaft, ω is the magnitude of the angular velocity of the input shaft, t is the phase shift between the input and output, and R is the gear ratio of the cycloidal drive, which is used to convert from input angular velocity to output angular velocity. Four phase shift measurements were taken for each drive after each hour interval to decrease the uncertainty in the measurement. The phase shift for each sample drive was calculated from the data for input and output positions versus time via cross-correlation.

The input and output shaft positions were recorded via two Vernier Rotary Motion Sensors, one connected directly to the output shaft of the sample drive and one connected to the input shaft via a belt. The Vernier Rotary Motion Sensors have an angular resolution of 0.25° (0.0044 rad) when measuring rotation below 7.25 rev/s, and the input shaft was always driven at 5 rev/s. Figure 4 contains an image of the measurement apparatus. The sensors were always zeroed before any measurement was made.

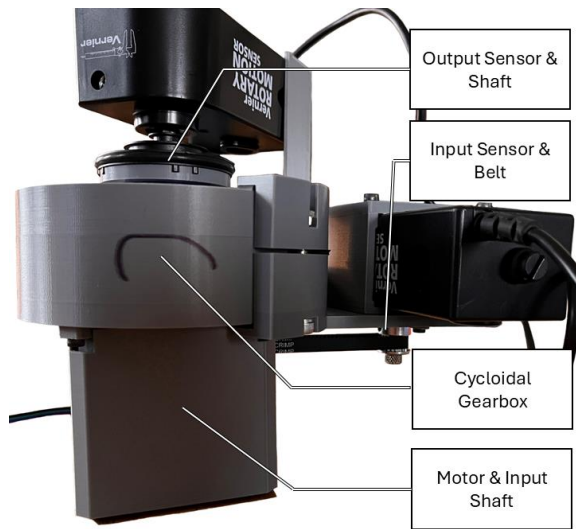


Figure 4: The cycloidal drive backlash measurement apparatus connected to the control sample drive. A NEMA-17 stepper motor is connected to the gearbox via the input shaft, which is also connected to the input rotary motion sensor via a belt. The drive output is connected directly to its sensor via an adapter.

The limiting factor for the precision of the measurements was the sampling rate of the sensors. The recommended sampling rate for the sensors is 100 samples per second, and they were run at 1000 samples per second when taking these measurements. Even at this increased rate, the uncertainty caused by the 1 millisecond sample spacing is significant.

WEAR-IN OF SAMPLE DRIVES

This study is only concerned with even wear distributions, and thus special consideration must be given to avoid uneven wear distributions and ensure positionally-independent backlash. Therefore, the wear-in process must be designed to ensure that no preference is given to any one portion of a drive's angular range. This was done by defining the wear-in cycle of each drive via a ' t value.' Each sample drive was assigned a unique t value, and over the course of the wear-in period, the output shaft of each drive spent $2t$ seconds rotating clockwise, then t seconds rotating counterclockwise, and so on. This prevented the uneven accumulation of wear, because the drives changed directions while in many angular positions. The t values of each sample drive are as follows: infinity, 12, 9, 6, and 3. A t value of infinity corresponds to the control drive, which underwent no direction changes over the wear-in period. These t values can be converted to a total number of direction changes over the 5 hour wear in period via unit conversion. The previously mentioned t values correspond

to 0, 1000, 1333, 2000, and 4000 total direction changes, respectively.

The sample drives were worn in two at a time, with the fifth drive worn in on its own. The wear in was done in one hour long increments, which allowed for the measurement of each drive's backlash after every hour of wear. Each drive was connected to a Twotrees NEMA-17 1.5 A stepper motor. The stepper motors were controlled via an Arduino Nano connected to two HiLetgo A4988 stepper drivers, which were themselves connected to the stepper motors of the two sample drives being worn in. The circuitry was assembled on a breadboard with a 1 A maximum rated current, and thus both drivers had their current limits set to 0.75 A to avoid damaging the board. The Arduino was coded such that each drive would independently undergo direction changes according to its t value. If, in future research, more drives are worn in simultaneously, this method becomes impractical and a different arrangement would be required. An image of the circuit is provided in Figure 5.

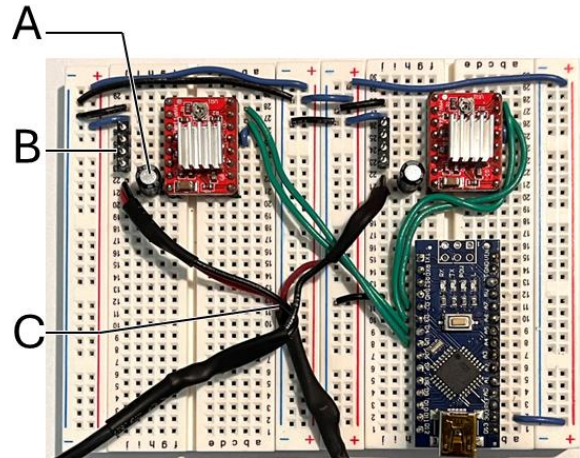


Figure 5: Circuit board used to control and power cycloidal drive wear-in. The Arduino Nano (bottom circuit board) is connected to a computer during wear-in for power. 12 VDC is delivered to the stepper drivers (red circuit boards) via the cables labelled C, and 100 μ F capacitors (labelled A) are placed across the power delivery positive and negative terminals to prevent power spikes from damaging the stepper drivers. The stepper motors (not pictured) are connected to each stepper driver via the risers labelled B, and control signals from the Arduino are sent to the stepper drivers via the green wires.

RESULTS AND DISCUSSION

The use of PLA cycloidal drives is a tempting choice for many hobbyists because of their high positional precision and ease of manufacture. This only holds true,

however, if PLA cycloidal drives can maintain positional precision. The data collected can help determine whether this is the case.

For every sample drive that was tested, and after every hour of elapsed wear-in time, four position versus time measurements were taken. For each 10 second long measurement, the position of the input and output shafts was recorded when given a triangular wave input signal. The input and output position data were then cross-correlated to determine the phase shift measurement, as seen in Figure 6.

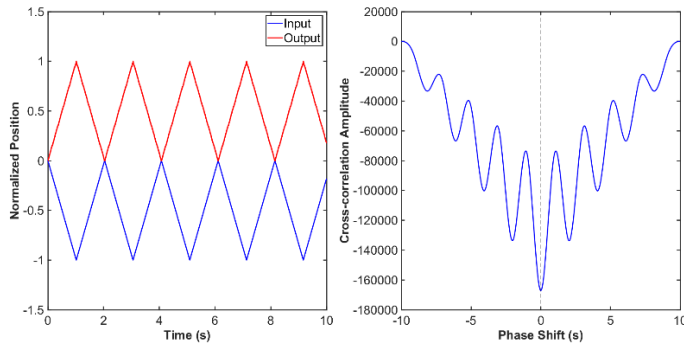


Figure 6: Normalized angular position for the input and output versus time (left), and cross-correlation of the two position signals (right). This set of plots is from one measurement of the control drive after five hours of wear. The dashed vertical line in the correlation graph is at the phase shift where the correlation has the largest magnitude. In the event of measurable backlash, the red output position data would be out of phase with the blue input position data. Accounting for all four measurements made for this drive in this wear-in state, the time shift between the input and output is 0.000 ± 0.001 seconds, where the uncertainty is limited by the sample rate of the sensors, 1 kHz.

The control drive, which underwent no direction changes over the course of the study, had a backlash of 0.000 ± 0.002 rad at every measured point in time, as seen in Figure 7. Every datum collected for this drive was a phase shift of 0.000 seconds, and the uncertainty in the final measured backlash is sourced from the sample rate of the sensors, 1 kHz. This is a null result, and thus shows that there is no measurable relationship between backlash and usage time over the course of 5 hours. It is likely that such a relationship does exist, but it does not have a measurable effect over a 5 hour period.

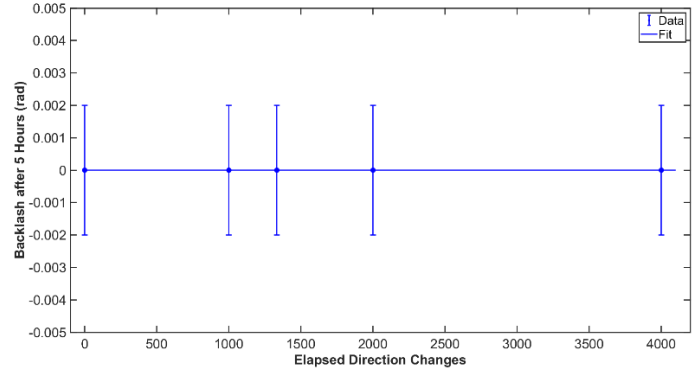


Figure 7: Backlash in control drive versus elapsed wear-in time, with curve fit. At every measured point in time, the backlash in the control drive was 0.000 ± 0.002 rad, and the resulting curve fit is a constant backlash of 0.000 ± 0.002 rad. This uncertainty was derived from the sampling rate of the sensors, 1 kHz. This indicates that, for PLA cycloidal drives, 5 hours of continuous use is not enough to cause a measurable increase in drive backlash.

Notably, this result contradicts that of the study done by Sensinger and Lipsey, because the cycloidal drives they tested had backlashes ranging from 0.4° to 1.4° (0.007 to 0.0024 rad) [6]. The cycloidal drives examined in the Sensinger and Lipsey study all had lower outer diameters than the drives in this study [6]. The largest disk outer diameter in Sensinger and Lipsey was 70 mm, and in this study the drives all have a disk outer diameter of 74.5 mm [6]. Every drive in Sensinger and Lipsey also had much higher gear reduction ratios than the drive design in this study; the 70 mm outer diameter drives in Sensinger and Lipsey had gear ratios in the range of 50 – 160, whereas the drive design in this study has a gear ratio of 20 [6]. These differences combined indicate that all necessary features for the function of a cycloidal drive, namely the lobes on the cycloidal disks and the diameters of the rollers, are much larger in the design used for this study. It is noteworthy that the cycloidal drives manufactured in the Sensinger and Lipsey study were not used to determine backlash, and the backlash figures they present were determined via simulation [6]. Sensinger and Lipsey also did not investigate any changes in drive backlash over time [6]. Because manufacturing techniques have no influence on simulated results, these results warrant further examination into the influence of feature scale and corresponding design decisions on drive backlash.

In the remaining PLA sample drives, the backlash measurements were similar to those of the control drive. Every sample drive had a measured phase shift of

0.000 seconds with uncertainty sourced from the 1 kHz sample rate of the sensors, resulting in a final backlash measurement of 0.000 ± 0.002 rad for every drive at every point in time. The final backlashes of each drive are shown in Figure 8, and the intermediate backlashes of each drive during the wear in period follow the trend from the control drive, namely that there is no change.

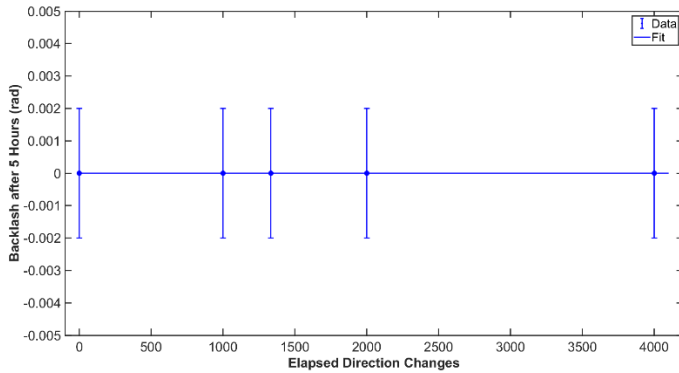


Figure 8: Backlash in all drives after 5 hours versus total direction changes each drive underwent. After the full 5 hour wear in period, the backlash in every drive was 0.000 ± 0.002 rad. This uncertainty was derived from the 1 kHz sampling rate of the sensors. This indicates that, over a 5 hour period, there is no measurable dependence between backlash and elapsed direction changes over said period.

The combined data for the backlashes in all drives after 5 hours indicates, once again, a null result. Therefore, PLA cycloidal drive backlash is not measurably dependent on the elapsed direction changes over a 5 hour wear-in period. Although there is likely some dependence between backlash in the drives and wear-in time, there is not necessarily a dependence between backlash and elapsed direction changes over that time. Therefore, it is entirely possible that future examination of cycloidal drive backlash will produce similar results.

There are a few limitations of this study to discuss. Primarily, the sensors used in this study only allow for sampling rates that are too low to detect any backlash in the sample drives. The recommended maximum sampling rate for the Vernier Rotary Motion Sensors is 100 Hz. For this study, they were run at 1 kHz, and this rate was still insufficient to measure any backlash in the drives. For future investigations of cycloidal drive backlash, sensors that allow for higher sampling rates should be used. It is also possible that backlash in cycloidal drives manifests differently than in standard gear-based reduction methods. In standard gear reduction systems, an output shaft cannot rotate before the input shaft completes rotation through the

backlash period, but in the cycloidal drives examined in this study it is entirely possible that backlash acts more as a ‘false set.’ The cycloidal drive outputs may rotate with the input shaft free of backlash until put under sufficient resistive load, in which case the output is forced to rotate through the drive’s backlash period. Therefore, it is recommended that further examination of cycloidal drive backlash introduces rotational resistance to the output shaft.

CONCLUSIONS

It was found that, for PLA cycloidal drives made via FDM 3D printing, backlash is lower than could be measured for the entirety of a 5 hour wear-in period, where it was measured at a constant 0.000 ± 0.002 rad at all times, regardless of the total number of direction changes undergone during that period. This shows that PLA cycloidal drives can maintain their initial positional precision for a period of 5 hours, regardless of usage pattern. This result is beneficial for the hobbyist robotics space, because it provides empirical information useful for informing decisions about the application of and material selection for cycloidal drives.

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